Candidate Seat no............

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (IT) Examination

March-April 2018

MA301.01/IT304/IT304.01 Modeling Simulation & Operational Research

Date: 05.04.2018, Thursday **Time: 01.30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

SECTION – I

Q - 1 Do as directed:

a. An optimization model **[1]**
A. Provides the best decision B. Provides decision within its limited context
C. Helps in evaluating various alternatives D. All of the above

b. What do you mean by Operation Research? List out phases of Operation Research **[2]**

c. Define condition on which Simplex method maximization problem terminates. **[1]**

d. When total supply is equal to total demand in a transportation problem, the problem is said to be **[1]**
A. Balanced B. Unbalanced
C. Degenerate D. None of the above

e. Define Decision variables and Objective function **[1]**

f. Students arrive at the help center of Charusat according to a Poisson input process with a mean rate of 40 per hour. The time required to serve a student has an exponential distribution with a mean of 50 per hour. Assume that the students are served by a single individual, find the average waiting time of a student. **[2]**

Q-2 Do as directed

a. A firm manufactures two types of products A and B and sells them at a profit of Rs. 2 on type A and Rs. 3 on type B. Each product is processed on two machines G and H. Type A requires 1 minute of processing time on G and 2 minutes on H; type B requires 1 minute on G and 1 minute on H. The machine G is available for not more than 6 hours 40 minutes while machine H is available for 10 hours during any working day. Formulate the problem as a linear programming problem. **[04]**

	Type of products (minutes)		
Machine	Type A (x_1 units)	Type B (x_2 units)	Available time (mins)
G	1	1	400
H	2	1	600
Profit per unit	Rs. 2	Rs. 3	

b Write the dual of the linear programming problem given below: **[03]**
Min Z = 2x2 + 5x3
Subject to:
$x1+x2 \geq 2$
$2x1+x2+6x3 \leq 6$
$x1 - x2 +3x3 = 4$ where $x1, x2, x3 \geq 0$

OR

b Max $Z = 80x1 + 55x2$ **[03]**
Subject to
$4x1 + 2x2 \leq 40$
$2x1 + 4x2 \leq 32$ Where $x1 \geq 0$, $x2 \geq 0$
Solve this linear programing problem using graphical solution

c Solve this linear equation using graphically and check it has multiple optimal solution or No optimal solution or Unbounded solution **[04]**
Max $Z = 3x1 + 2x2$
Subject to
$x1 + x2 \leq 1$
$x1 + x2 \geq 3$ Where: $x1 \geq 0$, $x2 \geq 0$

d Determine the initial basic feasible solution using Row Minima Method **[06]**

	W1	**W2**	**W3**	**W4**	**Availability**
F1	19	30	50	10	**7**
F2	70	30	40	60	**9**
F3	40	80	70	20	**18**
Req.	**5**	**8**	**7**	**14**	

Q - 3 Answer any TWO. **[10]**

a. A department head has four subordinates and four tasks have to be performed. Subordinates differ in efficiency and tasks differ in their intrinsic difficulty. Time each man would take to perform each task is given in the effectiveness matrix. How the tasks should be allocated to each person so as to minimize the total man-hours?

Task	**I**	**II**	**III**	**IV**
A	8	26	17	11
B	13	28	4	26
C	38	19	18	15
D	19	26	24	10

b. Solve the following Linear Programming Problem using Simplex method:
Maximize $z = 3x1 + 2x2$
subject to
$-x1 + 2x2 \leq 4$
$3x1 + 2x2 \leq 14$
$x1 - x2 \leq 3$ Where: $x1, x2 \geq 0$

c. Solve following linear programming problem using Two-Phase Method.
Max $Z = -3x1 + x2 - 2x3$
Subject to:
$x1 + 3x2 + x3 \leq 5$
$2x1 - x2 + x3 \geq 2$
$4x1 + 3x2 - 2x3 = 5$ where $x1, x2, x3 \geq 0$

SECTION – II

Q - 4 Do as directed:

a. To find out Initial Basic Feasible Solution which method is best? **[01]**

b. List out applications of simulation? **[01]**

c. Write some real life applications of CPM and PERT. **[01]**

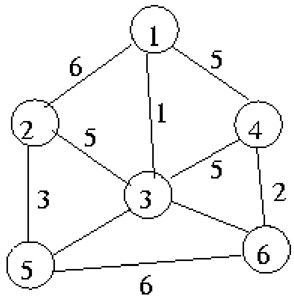
d. Explain terms: (a) Collusion (b) Jockeying **[01]**

e. List out difference between dijkstra's and floyd's algorithm **[03]**

Q-5 Do as directed

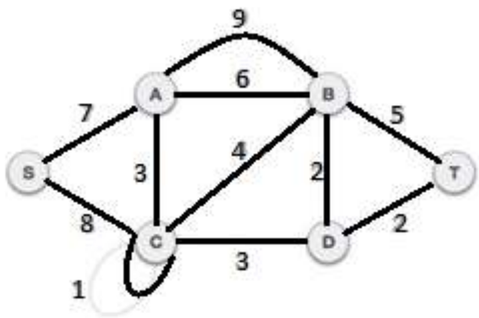
a Explain term M/M/1 (∞/FIFO) in Queuing System? **[02]**

b Apply Prim's algorithm and find minimum cost spanning tree for following Graph **[07]**



OR

b Apply Kruskal's algorithm to find the minimum cost spanning tree **[07]**



c At Bharat petrol pump, customers arrive according to a Poisson process with an average time of 5 minutes between arrivals. The service time is exponentially distributed with mean time = 2 minutes. On the basis of this information, find out **[05]**

1) What would be the average queue length?

2) What would be the average number of customers in the queuing system?

3) What is the average time spent by a car in the petrol pump?

OR

c Write the advantage and disadvantage of simulation. **[05]**

Q – 6 Attempt any Two: **[14]**

a. Chhabra Saree Emporium has a single cashier. During the rush hours, customers arrive at the rate of 10 per hour. The average number of customers that can be processed by the cashier is 12 per hour. On the basis of this information, find the following:

-Probability that the cashier is idle

-Average number of customers in the queuing system

-Average time a customer spends in the system

-Average number of customers in the queue

-Average time a customer spends in the queue

b. Consider the case of a dealer of a certain product for which the probability distribution of daily demand and the probability distribution of the lead time , both developed empirically by observations made over a long span of period, are as follows:

Probability distribution of daily demand :

Units Demand	3	4	5	6	7	8	9	10	11	12
Prob.	0.02	0.08	0.11	0.16	0.19	0.13	0.10	0.08	0.07	0.06

Probability distribution of lead time :

Lead time(days)	2	3	4	5
Prob.	0.20	0.30	0.35	0.15

The ordering cost is: 80 Rs.
Holding Cost per Unit is: 2 Rs.
Loss in profit per Unit per day is: 20 Rs.
Re-order Quantity: 40 Units
Re-order level 20 Units
Random number for daily demand: 68,13,09,20,73
Random number for Lead time : 47,74,85,25,68
Simulate for next 5 days and calculate Total Ordering cost and Holding cost.

c. Explain dummy activity in CPM with example and with neat sketch

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